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Research on the impact of matched effects between green advertising appeals and product type on consumer purchase intention

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ABSTRACT

With increasing global attention to environmental issues, green marketing has become a vital component of corporate strategy. Consequently, designing effective green advertising strategies to attract consumers has garnered significant interest. This paper employs experimental methods to investigate the impact mechanism of green advertising appeals—specifically, egoistic appeals versus altruistic appeals—on consumers' green purchase intentions. Additionally, it utilizes the Elaboration Likelihood Model and Construal Level Theory to elucidate the internal mechanisms at play. The findings indicate that: (a) green advertising appeals exert a substantial influence on consumers' green purchase intentions and perceived information credibility, with egoistic appeals exhibiting a more pronounced effect than altruistic appeals; (b) there exists a significant mediating effect through consumers' perceived information credibility; (c) product type serves as a moderator between green advertising appeals and consumers' purchase intentions. Specifically, when purchasing utilitarian products, egoistic appeals enhance consumers' green purchase intentions and perceived information credibility compared to altruistic appeals. Conversely, when purchasing hedonic products, altruistic appeals prove more effective than egoistic appeals. Ultimately, the study's conclusion offers insights and recommendations for enhancing the efficacy of enterprise green advertising strategies.

1. Introduction

More and more companies are embarking on a green transformation, actively producing and promoting green products that meet consumers' needs and enhance ecological and environmental benefits (Pinkse and Bohnsack, 2021). For example, Nike has launched its core ecological program, the 'Move to Zero' initiative, to address climate challenges, including the phasing out of single-use plastics at all Nike campuses, and to promote an important dialogue in the global design community through innovative circular design concepts. Not only do these green products contribute to positive environmental externalities, but they also bring many benefits to consumers, such as high-quality eco-detergents that provide deeper cleaning and better care, extending the life of clothes and reducing the environmental impact of wash water.

However, due to information asymmetry and a lack of environmental awareness, have led to a gap in consumer perception of green products, with one study noting that consumers often find it difficult to process information about the sustainability of a product, which leads them to shift from interest to skepticism (Luan et al., 2022). Therefore, companies often communicate green product information to consumers and the public by conducting green advertising campaigns to help consumers make decisions. Green advertising is an important part of corporate green marketing, which is not only a new way for companies to adapt to the development of society to promote their products, but also an important way to enhance consumer awareness and promote green consumption. Effective green advertising strategies can promote consumers' green purchasing and expand green market share for enterprises, realizing the environmental benefits of corporate strategy and gaining competitive advantages and opportunities (Sun et al., 2020). Therefore, companies have been cleverly designing their advertising messages to improve the effectiveness of their green advertising strategies (Knight et al., 2021), and some studies have shown that the focus of corporate green communication is increasingly on the altruistic environmental appeals and the egoistic health appeals of green products

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(Amron, A., 2018). These two appeals have been shown to have research value when combined with livestreaming in previous studies (Asante et al., 2023), and in addition, a study using VIRTUAL influencers revealed the difference between altruistic and egoistic appeals in influencing consumers' purchase intention (You and Liu, 2024).

In addition, given that green products have many different attributes, such as both practical functional attributes (health, safety, efficiency, and cost-saving) that are beneficial to consumers and sustainable attributes (low pollution, energy saving, and recyclability) that are beneficial to the ecological environment(Ashoush and Kortam, 2022). Consumers differ in their motivations for purchasing products and may seek the practical value of a product being healthy and safe, or the hedonic value of a product being symbolic and sustainable, and therefore focus on product information differently. Prior research has explored the relationship between hedonic products and trust (Lee and Park, 2023) and the combination with subjective busyness (Dai and Sheng, 2022). So how do product types (utilitarian and hedonic) combine with advertising appeals (altruistic or egoistic) to maximize consumers' purchase intentions? This is a topic worth exploring. In examining this issue, perceived credibility is crucial to consumer purchase decisions. Previous studies have shown a relationship with consumer purchase intention (J. M. Alcántara-Pilar et al., 2024; E. Rosillo-Díaz et al., 2024). With the help of the Construal Level Theory, we explore whether perceived credibility is an effective mediator in influencing consumers' purchase intentions.

To this end, this study aims to analyze the effect of altruistic and egoistic appeals as independent variables and product type as a moderating variable in green advertising appeals on consumers' green purchase intentions. At the same time, the study will also assess whether consumers' perceived credibility plays a mediating role in this process, thus providing a more comprehensive perspective for understanding the consumer decision-making process. Through this multi-dimensional analysis, we expect to provide theoretical support and practical guidance for the formulation of corporate marketing strategies.

2. Theoretical review and hypothesis development

2.1. Classification of green advertising appeals

Green advertising appeal is that enterprises introduce the green attributes of products through advertising, convey the values of health, environmental protection, and green, stimulate consumers' green demand, and promote the idea of green purchases. According to the collation of previous studies, scholars at home and abroad classify green advertising appeals from different perspectives (see Table 1). As green consumption is in the development stage, consumers' green knowledge reserve is limited. When purchasing green products, they are more inclined to consider the relevance of the product to their interests, thus generating either strong or weak purchase motives. The classification of egoistic and altruistic appeals elaborates the relevance of green products to their interests, which is easier for consumers to accept and understand, and the use of this classification to study the effectiveness of green advertising appeals has more theoretical and practical significance.

The effectiveness of altruistic and egoistic appeals is still controversial among scholars in previous studies. One study shows that consumers are usually more concerned about their interests and are relatively insensitive to the ecological and environmental impacts of products(Peattie, K., 2001). Altruistic advertisements may raise doubts about the quality of the product, whereas egoistic advertisements are more likely to win consumers' favor. However, Webb et al. (2007) argue differently, suggesting that although consumers' choice of green products implies giving up some personal benefits, they are more likely to engage in purchasing behavior when advertising appeals can be combined with environmental benefits.

Table 1
Review of articles in green advertising appeals.

Article	Classification	Perspective	Concept
Iyer (1993) Ku et al. 2016	Green appeal & non-green appeal	The green level of the product	Green advertising appeals emphasize the green attributes of a product or service related to environmental friendliness, whereas non-green advertising appeals only focus on attributes related to the consumer.
Kareklas (2014) Song et al. (2019)	Egoistic appeal & altruistic appeal	The perspective of purchase motivation	Egoistic advertising appeals focus on product attributes such as health, safety, and price, which are closely related to individual consumer benefits; whereas altruistic advertising appeals concentrate on the environmental benefits and eco-friendly aspects of products, which are closely linked to social and ecological values.
Yang et al. (2015)	Specific appeal&abstract appeal	Degree of specificity of information	Concrete demands refer to describing products with detailed information and specific methods; abstract demands use vague terms without explicitly stating product features.
Chan et al. (1996) Matthes et al. (2013)	Rational appeal &emotional appeal	The emotional perspective	Rational advertising conveys the quality, functionality, benefits, value, and price of a product, winning consumer trust and recognition through reasonable and evidence-based methods; emotional advertising focuses on appealing to consumers' emotions and feelings, inducing purchasing desires through emotional resonance and emotional stimulation.
Muralidharan et al. (2017)	Collectivist appeal &individualist appeal	Subject differences	Collectivist appeals emphasize group interests and unity, using group pressure to encourage consumers to follow environmental and social norms; individualist appeals focus on the benefits and value that products can bring to individuals, emphasizing personal goals.

In view of the above analysis and to gain insight into how different green advertising appeals will impact our research context, this study follows the categories of egoistic and altruistic appeals in green advertising appeals and further explores how these two types of appeals will affect consumers' willingness to purchase green products.

2.2. Green advertising appeal and consumer purchase intention

In green advertising messages, whether they are egoistic or altruistic appeals, their impact on consumers' purchase intention can be explained by the Elaboration Likelihood Model (ELM). The model illustrates the different paths of information processing, thus helping us to understand how consumers, after being exposed to green advertisements, form an identification with green products or services and ultimately influence their purchase decisions. The model elucidates different pathways of information processing, thereby helping us understand how consumers form identification with green products or services after exposure to green advertising and ultimately influence their purchasing decisions. This model posits that individuals engage in two distinct processing routes when encountering information: the central route and the peripheral route, which subsequently shape their attitudes or inclinations. The choice between these routes hinges on an individual's motivation and cognitive capacity to engage with the information. When individuals possess strong motivation and ample analytical abilities, they tend to process and decide via the central route. Conversely, when motivation and cognitive capacity are lacking, they resort to the peripheral route for processing and decision-making.

Egoistic advertising appeals prioritize consumers' personal interests, aiming to fulfill their desire for healthy and affordable green products. Consequently, consumers tend to engage in central route processing, meticulously evaluating information content through rational responses. In such instances, consumers gather persuasive product-related information, leveraging their professional knowledge and experiences to analyze it. This process culminates in the formation of psychological perceptions of the advertisement, ultimately leading to corresponding attitudes and behavioral intentions. Conversely, altruistic appeals underscore the environmental benefits of green products, highlighting their contribution to long-term sustainable development (Casalegno et al., 2022) and implying a social status signa (Chan, K. K., 1996). These appeals resonate with current cultural values centered on environmental protection but may not directly relate to individuals. Consequently, emotional responses triggered by the information content activate consumers' peripheral route processing, where heuristic clues and emotional reactions in advertisements play a pivotal role in shaping attitudes. In this scenario, consumers are drawn to the emotional satisfaction and image-building aspects of altruistic appeals, leading to positive attitude experiences and motivating green product purchases to fulfill consumer needs.

Existing studies have shown that, compared to altruistic appeals that process information through peripheral routes, egoistic appeals lead consumers to process information through a central route in a more detailed manner, resulting in more rational thinking and judgment, and consumers perceive higher functional attributes obtained (Ni et al., 2022). In essence, altruistic appeals typically evoke emotional responses and are processed via the edge path, potentially yielding corresponding advertising effects but susceptible to situational influences. Conversely, egoistic appeals prompt consumers to utilize the central route for information processing, aligning closely with their interests and prompting careful consideration and judgment. Thus, emphasizing egoistic demands related to consumers' interests and functional performance can enhance green product purchase intentions(Jäger and Weber, 2020). Based on the aforementioned analysis, the following hypothesis is posited:

H1. Different types of green advertising appeals significantly influence consumers' green purchase intentions. Specifically, compared to altruistic advertising appeals, egoistic advertising appeals exert a stronger impact on consumers' green purchase intentions.

2.3. The mediating role of perceived information credibility

According to Construal Level Theory (CLT), individuals employ concrete (low-level) and abstract (high-level) explanations to represent psychologically near and far events, respectively(Trope et al., 2010). When individuals perceive an event or object as psychologically distant, their thinking tends to become more abstract, focusing on a broader perspective(Trope et al., 2007). Conversely, when consumers perceive an event or object as psychologically close, their thinking becomes more concrete, leading them to adopt a narrower perspective and focus on the event in greater detail.

Egoistic appeals emphasize the practical benefits of a product, such as health, safety, and cost savings. These benefits are short-term and specific, closely tied to consumers' daily lives, thereby enhancing their knowledge and experience. In contrast, altruistic appeals highlight the environmental value of products, focusing on concepts like nature and environmental protection. However, these values are challenging to directly verify before or at the time of purchase. They possess long-term and abstract characteristics, making it difficult for consumers to make detailed judgments about their authenticity. Consequently, consumers are more inclined to view the green claims of enterprises as a new marketing tool. Therefore, egoistic advertising appeals tend to resonate more effectively with consumers. This resonance occurs because egoistic appeals align more closely with consumers' actual needs, leading to a heightened perception of authenticity. Conversely, altruistic advertising appeals emphasize environmental protection values that are harder to verify directly, resulting in a lower perception of authenticity among consumers.

Moreover, when consumers engage with advertising information, they tend to process egoistic appeals via the central path. Possessing strong motivation and information processing capabilities, they can effectively utilize knowledge and resources to evaluate advertising content, generate cognitive responses, and consequently develop a robust perception of authenticity. Conversely, altruistic appeals are often processed through the peripheral path, with consumers exhibiting weaker motivation for information processing. Only by evoking individuals' internal sense of environmental responsibility and green beliefs, and aligning advertising appeals with their inner emotions, can a favorable perception of advertising authenticity be fostered.

In the process of purchasing products, the credibility of information plays a pivotal role in consumer decision-making. Within the realm of green marketing, the level of trust in product information hinges on consumers' subjective perceptions, intertwining with product quality, service, and the sincerity and emotional resonance conveyed in advertising(Beverland and Farrelly, 2009). Perceived credibility is a subjective judgment made by consumers. Brands that emphasize attributes such as "nature" and "craftsmanship" in their product advertising information tend to foster cognitive and emotional engagement from consumers(Parkman and Krause, 2018). Moreover, when a product's narrative or information successfully aligns with consumers' needs and expectations, it contributes to their perception of its credibility(Petraglia J., 2009). Consumers' trust in product information stimulates their willingness to purchase, as it signifies their belief that the products adhere to described environmental standards. This, in turn, effectively alleviates consumers' sense of insecurity and uncertainty regarding product purchases, as they perceive the information provided as reliable and truthful. Enhancing consumers' perceived trust in the brand and its products trust is considered an effective strategy for establishing long-term relationships between the brand and consumers, thereby increasing consumers' willingness to purchase its products(Olk S., 2021). Building upon the aforementioned analysis, this study posits the following hypothesis:

H2. Compared to altruistic advertising appeals, egoistic advertising appeals exert a more positive impact on consumers' perceived information credibility and consumers' perceived information credibility serves as an intermediary between advertising appeals and green purchase intentions.

2.4. The moderating effect of product type

In this paper, adhering to the classification standard of product types outlined in Geuens et al., 2010's study, green products are categorized into two groups: utilitarian products and hedonic products, based on the individual consumer demands they aim to fulfill. Research indicates that the efficacy of distinct advertising appeal strategies varies depending on the type of green product being marketed(Gong and Wang, 2022). This divergence stems from differing consumer motivations for purchasing green products. While some consumers may be drawn to the functional performance of green products(Green and Peloza, 2014), others may be enticed by the symbolic significance associated with environmental consciousness, societal contribution, and elevated social status (Edinger-Schons et al., 2018; Gong et al., 2020). These varying purchase motives delineate consumers' demands and expectations regarding green products, with differing emphases on the product information they prioritiz(Jäger and Weber, 2020; Parkman and Krause, 2018). Hence, to achieve optimal persuasive outcomes, a crucial challenge we confront is how to effectively communicate product information to diverse consumer segments—namely, determining which type of appeal should be used. Given that consumers' responses to products may vary depending on the product type, previous studies have reached conflicting conclusions about the impact of different advertising appeals on the purchase intention of green products, which may be due to the various types of products involved.

Therefore, based on the classification of utilitarian and hedonic products, we explore how egoistic and altruistic advertising appeals can maximize consumer attraction under this classification.

Egoistic advertising appeals often emphasize green product attributes such as health, safety, and cost savings(Parkman and Krause, 2018). Consequently, consumers become more attuned to the health, organic, and other advantageous qualities of the product(Marcon, A., 2022), leading to higher functional and quality evaluations. However, they may overlook the environmental benefits of this process. This tendency aligns with the attribute characteristics typical of utilitarian products. When purchasing such products, consumers prioritize factors like quality and health. Thus, products that meet these functional expectations tend to elicit a stronger response(Petraglia J., 2009; Pinkse and Bohnsack, 2021). Compared to altruistic advertising appeals, egoistic appeals better cater to the needs of these consumers by highlighting product function and practical value. Consequently, the alignment between egoistic appeal and utilitarian benefits enhances consumer satisfaction.

Conversely, altruistic advertising appeals typically spotlight product attributes associated with environmental protection, recycling, and other sustainability aspects, emphasizing the role of purchasing green products in fostering long-term environmental and societal sustainability(Sun et al., 2021). When consumers encounter such advertisements, they tend to perceive the benefits of pollution reduction, carbon footprint minimization, and other environmental advantages, leading to a heightened evaluation of environmental efficacy. Functional quality considerations take a backseat, catering to the psychological needs of consumers who prioritize sustainable attributes and symbolic significance. According to Self-Consistency Theory, consumers select products that align with their self-expression needs, emphasizing brand symbolism and hedonic value(Suttikun et al., 2024), When products mirror their self-views, consumers experience a greater sense of self-image consistency(Trope, 2007). Given the emotional and symbolic attributes of hedonic products, altruistic advertising appeals that emphasize social benefits are more attuned to consumer needs for such products. When consumers purchase green products to champion environmental causes or uphold their positive self-concept, altruistic appeals that underscore pro-environment attributes prove more effective, evoking emotional resonance and eliciting a positive response from consumers.

Drawing from the preceding analysis and grounded in the Functional Consistency Theory and Self-Consistency Theory, we derive the following conclusions: For consumers invested in purchasing utilitarian products, their focus centers on the practical value encompassing product quality, safety, and price, driven by a pursuit of maximizing personal interests. Consequently, they exhibit a heightened receptiveness towards egoistic green advertisements, which emphasize the functional attributes of products, thereby fostering increased information credibility and purchase intention. In contrast, consumers seeking hedonic products prioritize maintaining a positive social image and contributing to environmental sustainability. They exhibit greater sensitivity towards altruistic green advertisements, which highlight the environmentally friendly aspects of products, resulting in elevated information credibility and purchase intention. The alignment between advertising appeals and the advertised products enhances the reliability of the appeals, rendering them most effective. Building upon these observations, this study posits the following hypotheses:

H3. Product type moderates the relationship between green advertising appeals and consumers' green purchase intention.

H3a. For utilitarian products, egoistic appeals exert a stronger positive influence on purchase intention compared to altruistic appeals.

H3b. For hedonic products, altruistic appeals exert a stronger positive influence on purchase intention compared to egoistic appeals.

H4. Product type moderates the relationship between green advertising appeals and consumers' perceived information credibility.

H4a. For utilitarian products, egoistic appeals exert a stronger positive influence on perceived information credibility than altruistic appeals.

H4b. For hedonic products, altruistic appealst exert a stronger positive influence on perceived information credibility than egoistic appeals.

To sum up, the theoretical model of this paper is shown in Fig. 1.

3. Research design and empirical analysis

3.1. Preliminary experiment

Before initiating the main study, a pilot test is carried out to validate the choice of suitable eco-friendly products and to minimize the potential bias that could arise from product type incongruity affecting the results. This preliminary phase is crucial for ensuring the robustness of the subsequent experimental findings.

Following the delineation of common green product types such as daily necessities, home appliances, and transportation(GE and Song, 2019), six green products with significant mentions in literature analysis and interviews were identified. In collaboration with guidance from college instructors, the products selected encompassed environmental-friendly laundry detergent, green cosmetics, energy-efficient table lamps, solar-powered watches, electronic bracelets, and new energy vehicles. Questionnaires were distributed randomly via the wenjuanxing platform, prompting participants to assess the practicality or hedonistic nature of each product. This facilitated the identification of the two products perceived as most representative by consumers.

The questionnaire begins by introducing the concepts of utilitarian products and hedonic products. Utilitarian products encompass products that fulfill essential daily or functional requirements, aiding individuals in practical tasks, and prioritizing functionality. On the other hand, hedonic products cater to consumers' sensory experiences, social values, and emotional needs, often imbuing greater symbolic value within the social circle. Subsequently, a Likert 7-level scale was

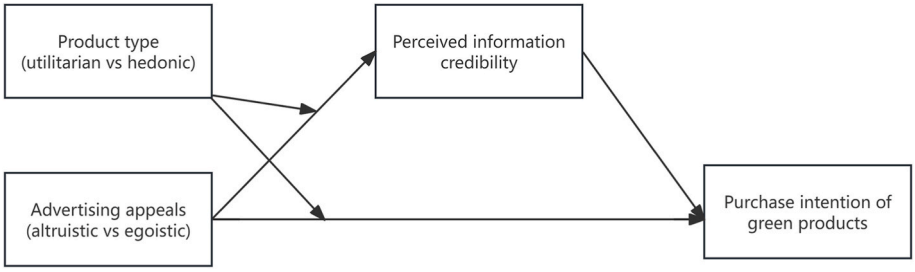


Fig. 1. Research model diagram.

employed to gauge the degree of categorization for these six products, with 1 indicating very inconsistent and 7 signifying very consistent. A total of 57 data were collected through the online questionnaire distributed by wenjuanxing, of which 50.88% (29) were male and 49.12% (28) were female; the age group was mainly concentrated in the age group of 19–25 years old, which accounted for 87.72% (50), and the age group of 26–30 years old, which accounted for 12.28% (7); and the education level was mainly concentrated in the undergraduate and postgraduate education. The utilitarian and hedonic degree of the six products is shown in Table 2.

Based on the statistical analysis, the environmental protection laundry detergent emerged as the utilitarian product, while the solar watch was identified as the hedonic product. Subsequent independent sample T-tests were conducted on the acquired data. The findings revealed that for the eco-friendly laundry detergent, the perceived practicality of the product (M practicality = 6.018, SD = 1.316) significantly outweighed the perceived hedonic degree (M hedonic = 3.088, SD = 1.845), $t(101) = 9.762$, $p < 0.001$. Conversely, for the solar watch, the perceived hedonic of the product (M hedonic = 5.333, SD = 1.673) significantly surpassed the degree of practicality (M practicality = 4.053, SD = 1.817), $t(112) = -3.915$, $p < 0.001$. These results affirm the successful manipulation of the green product type.

3.2. Experiment 1

To commence the principal research, a pilot experiment is undertaken to ascertain the suitability of selected green products and to prevent any bias stemming from mismatched product types that might skew the results. This initial step is essential for the validity and strength of the subsequent empirical results.

3.2.1. Experimental design

Based on the findings from the pre-experiment, green cosmetics with moderate utilitarian and hedonic attributes were chosen as stimulus products. In this study, for the selection of stimulus materials, we mimicked advertisement slogans that had been used by well-known brands in reality and adapted them to the experimental study by Yang et al. (2015). This choice aims to ensure that our experimental design can closely match real consumer scenarios, to more accurately simulate and assess the potential impact of advertising appeals on consumers' purchase decisions. We believe that through this close-to-reality research approach, we can more effectively reveal the mechanism of advertising strategies in the actual market environment and provide deeper insights into understanding consumer behavior.

Table 2
Statistical analysis of alternative products.

Product	Utilitarian level		Hedonic level	
	Mean	Standard deviation	Mean	Standard deviation
Eco-friendly laundry detergent	6.018	1.316	3.088	1.845
Energy-saving desk lamp	5.737	1.158	3.509	1.774
Electric vehicle	5.491	1.283	4.263	1.620
Smart bracelet	4.842	1.320	4.597	1.387
Green cosmetics	4.281	1.634	4.246	1.776
Solar-powered watch	4.053	1.817	5.333	1.673

The egoistic advertising appeal material emphasizes the product's benefits: "This cosmetics is crafted from natural and wholesome raw materials, containing concentrated essence that requires only half the usual amount for application. It's gentle on the skin and even suitable for sensitive skin types. The chic and recyclable packaging adds to its appeal. Moreover, its affordability promises long-term savings, with an additional 10% discount available for immediate purchase." In contrast, the altruistic advertising appeal material highlights the product's environmental contributions: "This cosmetic harnesses natural plant ingredients and ensures minimal environmental impact. Stringent controls during production mitigate the release of harmful substances. The eco-friendly packaging can be recycled, minimizing harm to the environment post-use."

The experiment employed a single-factor intergroup design involving two groups: one exposed to egoistic appeals and the other to altruistic appeals. A questionnaire survey method was utilized, with participants randomly assigned to either the egoistic or altruistic appeal group. In Experiment 1, 180 online questionnaires were distributed, yielding 166 valid responses. Each group consisted of 83 participants. Subjects were initially tasked with viewing the experimental materials. Subsequently, they answered manipulation questions regarding the two types of advertising appeals. The manipulation questions, adapted from Kareklas et al. (2014), assessed egoistic appeals with the prompt "The above advertisements are considered based on personal interests," The altruism in advertising was gauged by assessing whether the messaging was centered on environmental conservation principles. Following this, participants responded to measurement items regarding perceived information credibility and green purchase intention in succession. All responses were recorded on a Likert 7-level scale, where "1″ indicated strongly disagree and "7″ signified strongly agree. Perceived information credibility was measured using Lock and Seele' scale(2017), comprising five items assessing the accuracy, truthfulness, trustworthiness, lack of misleading information, and the reference value of the advertising content. Product purchase intention was evaluated using Lao' scale (2014), encompassing four items: willingness to learn more about the product, intention to purchase when necessary, readiness to recommend to friends and family, and preference for the product over similar alternatives.

3.2.2. Experimental results

It has been found that compared to other age groups, the younger group (people under 45 years old) pays more attention to green products and derived green advertisements. The younger group, benefiting from their higher level of education and open-mindedness, shows deep awareness and support for sustainability issues (Kuźniar et al., 2021), and with increased awareness of ecological issues, this group is increasingly inclined to choose green products in their consumer decision-making, thus reflecting a younger trend in relevant data reflecting the trend of youthfulness. Therefore, to ensure that we get real

and valid data results that can accurately reflect the green consumption psychology, the questionnaire distribution is more inclined to the younger group.

In Experiment 1, we distributed 180 questionnaires through the Chinese online platform wenjuanxing. Through screening, we excluded questionnaires that were incompletely filled out and those that took too short a time to fill out, because no incentive mechanism was implemented for participants during the experiment, and it is reasonable to speculate that incomplete filling out or too-fast submission may reflect a potential impact on the credibility of the results of the experiment; furthermore, considering that the validity of the study relies on the truthfulness and diversity of the responses provided by the respondents, and to ensure the accuracy of the study's conclusions, we excluded questionnaires with continuous consistent responses. After excluding the unqualified samples, 166 valid samples were finally collected: 83 samples from each of the egoistic and altruistic groups (see Table 3).

(a) Reliability test: In this experiment, Cronbach's alpha coefficient for perceived information reliability was 0.887, indicating strong internal consistency. Similarly, the Cronbach's alpha coefficient for the purchase intention measurement scale was 0.827, suggesting good reliability of the questionnaire.

(b) Manipulation check: The independent samples *t*-test for green advertising appeals revealed significant differences between groups. In the egoistic advertising group, participants perceived the emphasis on personal interest in the advertising information (M egoistic appeal = 5.759, SD = 1.590) to be significantly higher than the emphasis on environmental value (M altruistic appeal = 2.434, SD = 1.548), $t(164) = 13.656$, $p < 0.001$. Conversely, in the altruistic advertising group, participants perceived the environmental value emphasized in the advertising information (M altruistic appeal = 5.470, SD = 0.967) to be significantly higher than the emphasis on personal interest (M egoistic appeal = 3.205, SD = 1.552), $t(137) = -11.285$, $p < 0.001$. These results indicate successful manipulation of the green advertising appeals.

(c) Main Effect: To examine the impact of green advertising appeal on consumers' purchase intention, the experiment employs a single-factor analysis of variance between two groups (altruistic appeal and egoistic appeal). The results indicate significant differences between the two modes of green advertising that appeal to consumers' purchase intention. Compared to other appeal modes, green advertising utilizing egoistic appeal demonstrates a more favorable effect on purchase intention, with M egoistic appeal = 6.361, SD = 0.524, and M altruistic appeal = 5.611, SD = 0.650. The F-test $(1,164) = 66.943$, $P < 0.001$, confirming H1.

(d) Mediating Effect: One-way ANOVA on information credibility reveals significant differences between the two modes of green advertising appeal. The egoistic appeal in green advertising outperforms altruistic appeal in enhancing consumers' information credibility (M egoistic appeal = 6.306, SD = 0.722; M altruistic appeal = 5.439, SD = 0.582; F $(1,164) = 72.601$, $P < 0.001$), confirming H2.

Table 3
Demographic information.

Variable	Item	Sample Size
Gender	Male	69
	Female	97
Age	18–24	32
	25–34	40
	35–44	46
	45–54	28
	55 and above	20
Education	High School and below	33
	Bachelor	90
	Master and above	43

Subsequently, utilizing the PROCESS plug-in in SPSS, the Bootstrap method conducts intermediary testing, selecting Model 4 from the plug-in (5000 times sampling), with the result displayed in Fig. 2. Green advertising appeal significantly impacts information credibility (β = 0.545, SE = 0.134, 95%CI[0.164, 0.692], excluding 0). Information credibility positively influences consumers' green purchase intention (β = 0.677, SE = 0.057, 95%CI[0.490, 0.714], excluding 0), and the mediating effect of information credibility is significant (β = 0.258, SE = 0.120, 95%CI[0.029, 0.500], excluding 0). When the mediating effect is present, the direct effect interval does not contain 0 (95%CI[0.055, 0.445]), indicating that perceived information credibility plays a partially mediating role, thus verifying H2.

3.2.3. Summary

Experiment 1, by manipulating two types of green advertising appeals, demonstrated the impact of green advertising appeals (egoistic appeal vs altruistic appeal) on green purchase intention and the mediating role of perceived information credibility between them. Specifically, compared to other appeals, egoistic appeals enhance consumers' perceived information credibility and stimulate their green purchase intention. However, this experiment focused solely on a single product, neglecting the potential influence of product type on consumers' green purchase intention. To address this gap, Experiment 2 is designed in this study. It incorporates two different product types (utilitarian products vs hedonic products) to further tailor green advertising appeals, establish a stronger link between advertising appeals and products, and investigate the interplay between advertising appeals and product types.

3.3. Experiment 2

In Experiment 2, while enhancing the experimental materials, we altered the specific products and introduced product types (utilitarian products vs hedonic products) as moderating variables. This adjustment aimed to investigate how these distinct product types influence the impact of advertising appeals on green purchase intention and perceived information credibility.

3.3.1. Experimental design

Utilizing the data from the pilot phase, we determined that the eco-friendly laundry detergent, emphasizing environmental conservation, was a practical choice for green products, while the solar-powered watch was recognized as a hedonic green product.

Four experimental materials were developed, employing a 2 (green advertising appeal: egoistic appeal vs altruistic appeal) × 2 (product type: utilitarian products vs hedonic products) intergroup design. The design of the experimental materials was informed by Yang et al.'s previous research(2015) and tailored to our specific experimental context.

The egoistic advertising appeal for the eco-friendly laundry detergent emphasizes its skin-friendly attributes, utilizing natural plant ingredients that meet low allergen standards, ensuring safety even for sensitive skin. It boasts a 99% bactericidal effect and eliminates odors, effectively combating mites. Additionally, its concentrated formula means only a quarter of the usual amount is necessary. Furthermore, purchasers can enjoy a 10% discount if they buy now.

The altruistic advertising appeal for the eco-friendly laundry detergent highlights its use of pure natural degradable raw materials, primarily natural plant ingredients, devoid of acidic or toxic substances. The wastewater from washing can be safely used to irrigate flowers and plants without harming aquatic life, ensuring environmental safety. Moreover, it features low foam and easy rinsing properties, saving 50% water compared to regular laundry detergent. The detergent bottles are crafted from 100% recycled plastic, effectively cutting down carbon emissions. In the long term, using this detergent not only safeguards the ecological environment but also fosters sustainable development.

The egoistic advertising appeal for the solar watch highlights its

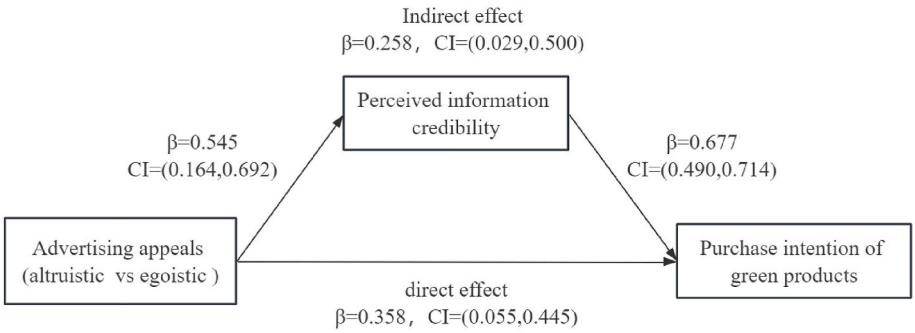


Fig. 2. Mediation analysis diagram.

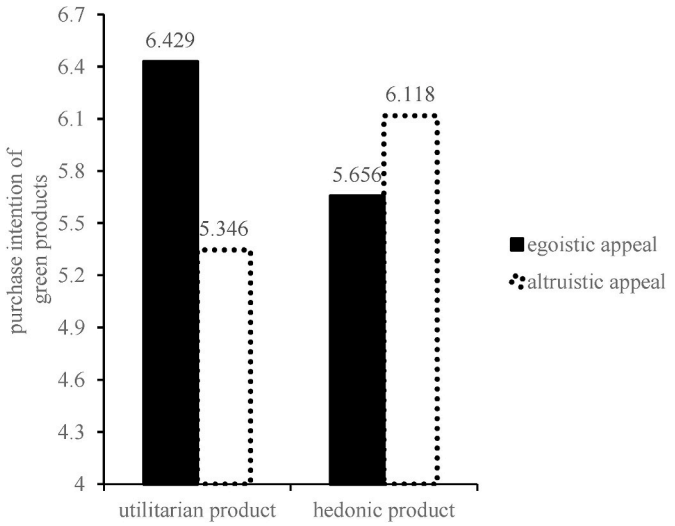


Fig. 3. The moderating effect of product type on advertising appeal and purchase intention.

versatility, affordability, and user-friendliness. It utilizes light energy drive technology, automatically absorbing light energy when worn daily, eliminating the need for battery charging or replacement. The watch ensures precise timekeeping through built-in system processing for automatic calibration, ensuring long-term stability. Additionally, purchasers can avail of a 10% discount if they buy now.

The altruistic advertising appeal for the solar watch emphasizes its environmental friendliness. It employs a light energy drive system, converting constant light sources into electricity through solar panels, thereby achieving zero resource loss. By eliminating the need for disposable batteries, it reduces pollution to soil, water, and other resources, while also fostering long-term sustainability and protecting the Earth's ecosystem.

Subjects were randomly assigned to one of four groups. For Experiment 2, 380 questionnaires were distributed online, yielding 363 valid responses. Based on combinations of green advertising appeals and product types, the questionnaires were divided into four sets: Questionnaire 1 (egoistic appeals × utilitarian products, 88 responses), Questionnaire 2 (altruistic appeals × utilitarian products, 91 responses), Questionnaire 3 (egoistic appeals × hedonic products, 93 responses), and Questionnaire 4 (altruistic appeals × hedonic products, 91 responses). During the experiment, participants first viewed the stimulus materials, followed by answering a series of questions in the questionnaire, consistent with Experiment 1.

3.3.2. *Experimental results*

Experiment 2 also distributed 380 questionnaires online through Wenjuanxing, and a total of 363 valid samples were harvested after the screening: 88 questionnaires for Questionnaire 1 (egoistic appeals × utilitarian products), 91 questionnaires for Questionnaire 2 (altruistic appeals × utilitarian products), 93 questionnaires for Questionnaire 3 (egoistic appeals × hedonic products), and 91 questionnaires for Questionnaire 4 (altruistic appeals × hedonic products), and the specific basic characteristics of the samples are shown in Table 4.

(a) Reliability test: In this experiment, Cronbach's coefficient alpha for perceived information reliability was 0.858, and for purchase intention, it was 0.827, indicating good questionnaire reliability.

(b) Maneuverability test: The independent sample *t*-test of green advertising appeal revealed significant differences. In the egoistic appeal group for utilitarian products, subjects perceived personal interest emphasized in the advertisement (M egoistic appeal = 5.625, SD = 0.732) significantly higher than environmental value (M altruistic appeal = 3.386, SD = 0.877), $t(174) = 18.387$, $p < 0.001$; Conversely, in the altruistic appeal group for utilitarian products, the environmental value emphasized in the advertisement was significantly higher (M altruistic appeal = 6.275, SD = 1.165) than personal interest (M egoistic appeal = 2.714, SD = 0.671), $t(180) = -25.261$, $p < 0.001$. In the hedonic egoism appeal group, subjects perceived personal interest emphasized in the advertisement (M egoistic appeal = 5.667, SD = 1.136) significantly higher than environmental value (M altruistic appeal = 3.011, SD = 0.759), $t(161) = 18.751$, $p < 0.001$; Similarly, in the altruistic appeal group for hedonic products, the environmental value emphasized in the advertisement was significantly higher (M altruistic appeal = 5.714, SD = 1.715) than personal interest (M egoistic appeal = 2.912, SD = 0.877), $t(134) = -13.879$, $p < 0.001$. These results indicate significant differences in advertising appeals across different product types, demonstrating successful manipulation of green advertising appeal materials.

(c) Adjustment effect: To examine the moderating effect of product type on green advertising appeal and consumers' purchase intention, ANOVA was conducted with green advertising appeal and product type as independent variables and purchase intention as the dependent variable. The results revealed a significant

Table 4
Demographic information.

Variable	Item	Sample Size
Gender	Male	182
	Female	181
Age	18–24	54
	25–34	80
	35–44	105
	45–54	69
	55 and above	55
Education	High School and below	59
	Bachelor	192
	Master and above	112

interaction term between green advertising appeal mode and product type ($F(1,359) = 70.419$, $p < 0.001$), thus confirming H3. Further simple effect analysis indicated that when the product type is utilitarian products, consumers' purchase intention for green products in the egoistic appeal group (M egoistic appeal = 6.429, SD = 0.434) is significantly higher than in the altruistic appeal group (M altruistic appeal = 5.346, SD = 0.869), $F(1,359) = 68.230$, $p < 0.001$, establishing H3a. Conversely, when the product type is hedonic, consumers' purchase intention for green products in the altruistic appeal group (M altruistic appeal = 6.118, SD = 1.046) is significantly higher than in the egoistic appeal group (M egoistic appeal = 5.656, SD = 1.006), $F(1,359) = 12.782$, $p < 0.001$. Therefore, H3b is supported, as depicted in Fig. 3.

To examine the moderating effect of product type on the relationship between green advertising appeal and information credibility, an ANOVA was conducted. Green advertising appeal and product type served as independent variables, while information credibility was the dependent variable. The analysis revealed a significant interaction between green advertising appeal and product type, $F(1,359) = 86.411$, $p < 0.001$, confirming H4. Further examination through simple effect analysis indicated that for utilitarian products, consumers exposed to egoistic appeals demonstrated significantly higher information credibility (M egoistic appeal = 6.084, SD = 0.513) compared to those exposed to altruistic appeals (M altruistic appeal = 4.813, SD = 0.581), $F(1,359) = 95.482$, $p < 0.001$, supporting H4a. Conversely, for hedonic products, consumers exposed to altruistic appeals exhibited significantly higher information credibility (M altruistic appeal = 5.365, SD = 1.132) than those exposed to egoistic appeals (M egoistic appeal = 4.938, SD = 1.058), $F(1,359) = 11.092$, $p = 0.001 < 0.01$, confirming H4b. These findings are illustrated in Fig. 4.

3.3.3. Summary

Experiment 2 enhanced the experimental materials, and the results demonstrated that the type of product could serve as a determining factor in the mechanism through which advertising demands influence green purchase intention. In particular, the type of product and advertising demands were found to interactively affect consumers' green purchase intention. When consumers purchase products as necessities, the egoistic appeal of the advertising information is more effective; whereas, when consumers buy products for pleasure, they tend to favor the altruistic appeal of the advertising information.

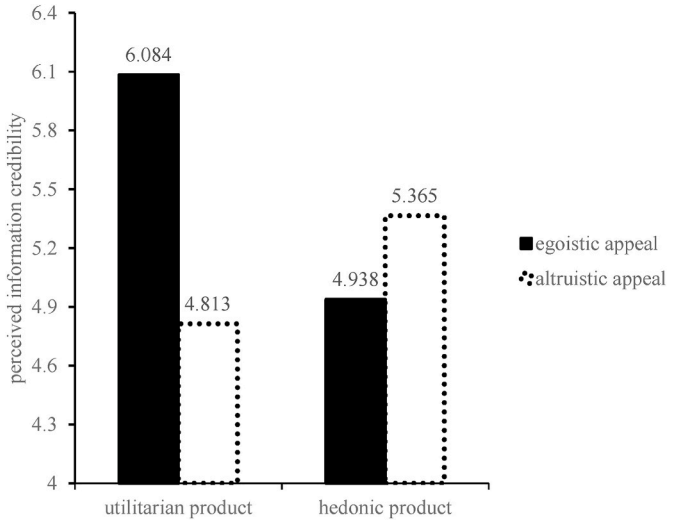


Fig. 4. The moderating effect of product type on advertising appeal and information credibility.

4. Research conclusions and implications

4.1. Research conclusions

Based on the sophisticated capabilities of the Elaboration Likelihood Model and the Construal Level Theory, this research has developed a conceptual framework that examines how green marketing messages can influence consumers' decisions to engage in eco-friendly buying behaviors. This framework explores the mechanisms through which green advertising can stimulate environmentally responsible consumer intentions.

The findings underscore a significant and positive correlation between green advertising appeals and consumers' inclination towards green purchases. Notably, egoistic appeals exhibit a stronger influence on consumers' green purchase intentions compared to altruistic appeals. Moreover, the impact of green advertising appeals on consumers' green purchase intentions is mediated by the perceived credibility of information. In the domain of consumer behavior towards environmental products, the influence of egoistic versus altrustic messages diverges based on the product type. For practical, physical items, egoistic messages more effectively strengthen consumers' intentions to buy eco-friendly options and enhance their perception of the reliability of the information provided. In contrast, for products that offer sensory or pleasure-based experiences, messages that evoke a sense of community or environmental stewardship are more likely to sway purchasing decisions. This distinction is crucial for crafting targeted marketing strategies that resonate with consumer motivations in the context of sustainable consumption.

4.2. Theoretical significance

(a) With little existing research delving into the mechanism linking green advertising appeal and consumers' purchase intentions, this research aims to address this gap by proposing the moderating role of product type in the persuasive pathway of green advertising. This study aims to evaluate the persuasive power of green promotional strategies by considering the influence of product-specific attributes. There is a spectrum of opinions among academics regarding the comparative effectiveness of appeals that are focused on the collective good versus those centered on individual benefits. It is posited that the success of these appeals fluctuates contingent on various contextual factors.

Given the distinct preferences of consumers for various products, the emphasis on advertising appeals should be adjusted accordingly when promoting green products with different attributes. Thus, this study categorizes products into utilitarian products and hedonic products based on their characteristics, investigating the influence of these two product types on diverse advertising appeals and consumers' green purchase intentions. This approach offers a fresh perspective for enterprises seeking to craft effective green advertisements.

(b) Explaining the persuasive effect of green advertising appeals from an innovative perspective. The persuasive effect of two green advertisement appeals, egoistic and altruistic, differs, which is affected by a variety of factors. Currently, there is insufficient research on consumers' perception of advertisement messages, and consumers' response to advertisements is a kind of mental processing process, while their level of perceived advertisement messages affects this mental processing process. Therefore, this study introduces consumers' perceived information credibility as a mediator and elaborates on the effectiveness of the two green advertisement appeals based on the Construal Level Theory, which expands the application of the Construal Level Theory in the field of green advertisement, and further enriches the

research on the mediating mechanism between green advertisement appeals and purchase intention.

4.3. Practical significance

With an increasing number of companies adopting sustainable practices and offering environmentally friendly products, there's a growing emphasis on encouraging consumers to embrace green purchasing behaviors, aiming to capitalize on strategic business advantages. In the fiercely competitive market landscape, enterprises must devise effective product and advertising strategies to captivate and motivate consumers' purchasing desires. Therefore, aligned with the research theme and conclusions, this section offers pertinent suggestions and insights for enterprises in the realm of green marketing, as outlined below:

To begin with, for green products whose categorization is challenging, employing altruistic advertising appeal strategies is advisable. This entails directing product advertising and marketing efforts toward highlighting functional value and individual benefits. Initially, advertising should underscore the safety and health advantages of green products. This approach not only caters to people's quality of life needs but also aligns with environmental protection and resource sustainability principles. Moreover, once a product is labeled as "green," consumers tend to perceive its "environmental protection attribute" as inherent(Long et al., 2024), albeit with difficulty in directly perceiving its environmental impact in the short term. Consequently, consumers prioritize maximizing their interests, focusing on the functional utility of green products. In marketing green products, enterprises should concentrate on their functional attributes and underscore the direct value they offer to consumers. Conducting market research enables a better understanding of consumers' specific functional expectations, allowing for the integration of these insights into product design and production practices to meet actual consumer demands. This, in turn, enhances the market competitiveness of green products. When designing green advertisements, the emphasis should be on showcasing the functional benefits that products bring to consumers, such as health, safety, and cost savings. By catering to consumer needs in this manner, enterprises can boost purchase willingness, thereby achieving the objectives of brand building, image promotion, and profitability.

Secondly, it's crucial to fine-tune the content of green advertising appeals and accentuate the attributes of green products to bolster consumer information credibility. To instill trust in advertising, enterprises should craft advertisements ensuring clarity, organization, and substantiation with ample evidence. In the realm of green advertising, alignment with reality is imperative. It's vital not to embellish or falsely tout the green or functional attributes of products for egoistic purposes. Instead, providing credible evidence to substantiate the green value of products, such as production material lists or eco-labels, is essential. Moreover, promoting green products through official platforms or employing reputable spokespersons with positive images can reinforce consumer confidence. This association between authentic advertising and trustworthy products can effectively stimulate purchase intentions. However, with the rise of green advertising, more and more greenwashing (a combination of the following tactics: Misleading or vague claims about the green credentials of an organization or product (Williams, J., 2024) and Growing demand for environmentally friendly products has led to an increase in companies exaggerating their environmental credentials (Dutta-Powell et al., 2023)). The emergence of behaviors may lead to advertising that deceives consumers and violates their right to information and safety. Companies may lose consumer trust due to greenwashing behaviors, affecting customer stickiness and brand loyalty, resulting in some losses, so regulators and public policymakers will also play an important role in how our research will advance practice. Therefore, regulators and public policymakers should do the following: (a) Strengthen the publicity and education for consumers and enterprises to enhance the public's ability to recognize greenwashing, and the regulators should formulate various green advertising and marketing standards, which can largely enhance consumers' knowledge of the products and prevent greenwashing to a certain extent, thus increasing the purchasing willingness and purchasing behaviour. And encourage the public to participate in the supervision; (b) advocate the concept of green advertising, promote environmentally friendly advertising production methods, through strengthening publicity and education, formulating standards and norms, and encouraging technological innovation, etc., to promote the green development of the advertising industry; (c) the government should improve the legislation, strengthen the constraints of the legal system and regulations, and make clear the legal consequences of the behavior of greenwashing, as well as to increase the penalties.

Thirdly, enterprises must tailor their advertising approaches to align with the unique attributes of their products, ensuring accurate delivery for enhanced credibility and persuasiveness in green advertising. Specifically, for utilitarian green products catering to consumers' functional needs, such as eco-friendly laundry detergent, recycled paper, and energy-efficient light bulbs, advertising design should prioritize egoistic information directly relevant to consumers' personal interests. This includes highlighting product health benefits, economic advantages, and dependable quality. Conversely, for hedonic green products like solar-powered watches, and electronic bracelets, which fulfill consumers' inner psychological and emotional desires for enjoyment and fulfillment, advertising should emphasize public value information that promotes overall benefits for society and others. This encompasses showcasing the environmental, ecological, and low-carbon attributes of green products, fostering a sense of collective responsibility and contribution to societal well-being.

4.4. Limitations and prospects

This paper concentrates on a novel product classification dimension, delving into the efficacy of green advertising appeals and scrutinizing their internal mechanisms, thereby expanding the practical domain of existing theories. In addition, in the manipulation of the experimental context, the green advertisement claims and product types were manipulated only through the form of textual stimuli, so that the test subjects imagined that there was a gap in the advertisement effects in the hypothetical context compared with the actual broadcast of video advertisements or offline supermarket shopping, which may limit the general applicability and external validity of the findings. Therefore, we suggest that future researchers adopt empirical evidence, such as field experiments, to methods to validate the findings of this study further to enhance the robustness and credibility of the conclusions.

Limitations in research methods: In this study, items commonly encountered in consumers' daily lives were selected based on relevant literature. However, whether the research findings apply to high-priced green products (such as luxury products, household appliances, automobiles, etc.) warrants further investigation. Additionally, in the experimental setup, only text-based stimuli were utilized to manipulate green advertising appeals and product types, prompting test subjects to imagine hypothetical scenarios. Compared to actual video advertisements or in-store shopping experiences, this approach may create a disparity in advertising effectiveness. There remains room for improvement in the selection of experimental materials and the design of stimuli effects. Therefore, future studies could consider incorporating high-priced green products such as luxury products and energy-efficient home appliances as experimental stimuli to validate the findings of this study. Furthermore, integrating real-life scenarios into experiments could be considered to mitigate potential errors in research results stemming from methodological limitations.

Examining the limitations of variables: Consumers' inclination towards green purchases is influenced by various internal and external factors. In this study, control variables solely encompass demographic characteristics. However, factors such as product price and brand also

significantly impact consumer behavior. Future research endeavors could incorporate additional control variables to gain a comprehensive understanding of the formation process of green purchase intentions. This study delves into how green advertising appeals, acting as external stimuli, influence individual cognition and subsequently trigger behavioral responses. Individual internal states comprise cognitive and emotional aspects, yet the mediating role of individual emotional states in this pathway is not considered, with the focus solely on the perceived trustworthiness of information, which falls within the realm of cognitive states. Hence, future research could delve deeper into the role of emotions, thereby comprehensively exploring the influence mechanism of green advertising appeals on purchase intentions.

Finally, the theoretical applications from the academic literature and terminology provided in this paper may help executives identify the issues they face as they seek to market more green advertising claims. Practicing marketing consultants and policymakers may also find the works cited in our paper helpful in bridging the gap between theory and practice; entrepreneurial decision-makers may also be able to create a competitive advantage by extending these insights.

CRediT authorship contribution statement

Fei Zhang: Writing – review & editing. **Yi Zhang:** Writing – original draft. **Shiyu Liao:** Investigation. **Xingjian Zhou:** Funding acquisition. **Xiaogang Ma:** Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

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Data availability

The data that has been used is confidential.

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